# Identifiability-Aware Hypothesis Generation for Undeciphered Scripts: Formal Claim Levels, Negative Controls, and a Sequence Model Anchored in the Real Rongorongo Corpus

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## Abstract

**Problem.** Statistical analyses of undeciphered scripts routinely slide from structural signal to semantic claims, a slide at the center of the Indus-script entropy debate. We ask what can and cannot be concluded from internal statistics alone, and build a system that operates strictly within those limits. **Method.** We formalize decipherment as an ill-posed inverse problem under a relabeling group action: the posterior over correspondences is constant on group orbits (T1), any decoder consuming invariant statistics obeys a Fano bound (T2), and we derive finite-sample concentration for windowed co-occurrence/PPMI spectra (T3), recovery conditions for anchored Procrustes alignment (T4), and the selection bias of language-model fusion decoding (P6). These results ground a three-level claim protocol — structural signal (C1), rankable hypothesis (C2), anchored semantics (C3) — as a theorem rather than a style guide (P7). **Results.** On the real Rongorongo corpus (tablets A–F, 5,460 glyphs, 941 Barthel types) we train a Transformer that maps Rapanui-like glosses to real Barthel codes and decode with beam search fused with a real-corpus trigram LM. Against three baselines and four ablations over six metrics with bootstrap CIs, the full system is the only one that is simultaneously non-degenerate (max repetition run 2), diverse (distinct-2 0.72) and within the realism band of held-out real lines. Permutation tests with BH correction confirm positional-bias signal in the Indus corpus (FDR $\leq 0.05$). **Limits.** The parallel corpus is synthetic by logical necessity (T1); all outputs are C2 hypotheses, and several spectral sanity checks fail on the real corpus — we report them.

## 1 Introduction

When Rao et al. reported entropic evidence for linguistic structure in the Indus script [1], Sproat and colleagues replied that the statistics in question could not, even in principle, establish what they were being read as establishing [2, 3]. The debate is often framed as a disagreement about corpora or estimators. We take a different view: it is a disagreement about *which statements internal statistics can decide at all* — and that question has a mathematical answer.

This paper develops the answer and builds a working system inside its boundaries. Our starting point is elementary but consequential: every statistic computed from an undeciphered corpus without external anchors is invariant under consistent relabeling of the signs, so no procedure consuming such statistics can distinguish a correspondence hypothesis from its relabelings (Theorem 3.1). What internal statistics *can* do is (i) falsify structural null models and (ii) rank hypothesis families. We freeze this into a three-level protocol — C1 structural signal, C2 rankable hypothesis, C3 anchored semantics — and prove that the C2/C3 boundary is exactly the boundary drawn by the group action (Proposition 3.7).

Within these limits we then go as far as the data allows. We integrate a real Rongorongo corpus (tablets A–F in Barthel transcription; 5,460 glyphs, 941 types) and build a translator from Rapanui-like glosses into *real* Barthel code sequences: a Transformer trained on a synthetic-by-necessity parallel corpus whose glyph side is anchored in real unigram/bigram statistics, decoded by beam search with shallow fusion of a trigram LM estimated on the real tablets. The system is evaluated against baselines and ablations on six metrics with bootstrap confidence intervals, with the selection bias of fusion decoding characterized *before* it can be mistaken for a discovery (Proposition 3.5).

**Contributions.**

1. A formal epistemology for computational decipherment: the C1/C2/C3 claim protocol, derived as a corollary of a group-theoretic non-identifiability theorem rather than posited as editorial practice (Section 3, D7/P7).
2. Finite-sample theory for the standard pipeline: Fano-type impossibility with an empirical-orbit refinement (T2), matrix-Bernstein concentration for windowed PPMI spectra with a Wedin transfer (T3), and noisy-anchor Procrustes recovery with a margin condition — whose instantiated threshold we test and report as vacuous on our benchmark (a documented gap

between sufficient theory and observed accuracy, T4).
3. A Rongorongo hypothesis engine anchored in the real corpus: class-restricted parallel-corpus construction declared as a C2 hypothesis, Transformer translation into attested Barthel codes, and fusion decoding whose selection bias is proved and then verified empirically by a $\lambda$-sweep (P6, Figure 2).
4. An evaluation protocol for generators over undeciphered scripts: six complementary metrics against real-corpus statistics, three baselines, four ablations, a contamination audit of the held-out split, permutation tests with FDR control, and a pre-registered blind-judge discrimination protocol (Section 6).

All numbers in this paper regenerate from seed-fixed scripts named in Section 10.

## 2 Related work

**Rongorongo.** The glyph inventory and numeric transcription system we use descend from Barthel [4]. Statistical approaches include Pozdniakov & Pozdniakov's sign-frequency comparisons with Rapanui phonotactics [5], Horley's allographic and statistical analyses [6], and Melka's studies of the Santiago Staff [7]; Guy identified calendrical structure on tablet Mamari [8]. Fischer's corpus and reading proposals [9] remain contested; we take no position on any reading claim — by Proposition 3.7 our pipeline cannot decide one. The transcription we consume (the RR corpus, phspaelti encoding of tablets A–F) embeds paleographic segmentation decisions that we inherit and flag as a limitation.

**Computational decipherment.** Snyder et al. deciphered Ugaritic against known-cognate Hebrew [10]; Luo et al. extended neural decipherment to Ugaritic and Linear B with minimum-cost flow [11]; Knight et al. treated decipherment as unsupervised learning over ciphers [12]. The structural difference to our setting is the anchor: those systems assume a known related language, which supplies exactly the external information $K$ that Theorem 3.2 prices. Without it, their objective would be orbit-invariant and their outputs C2 at best — which is where we deliberately operate.

**The entropy debate, engaged directly.** Sproat's critique [2, 3] of entropic arguments [1, 13] makes three points: (i) nonlinguistic symbol systems can produce the same statistics; (ii) results depend on corpus-preparation decisions; (iii) general-science venues under-review such claims. Our framework absorbs (i) as a theorem: conditional entropy and spectra are $G$-invariant, hence C1-only — they can falsify randomness nulls, never establish language (Definition 3.6). We answer (ii) operationally: every statistic ships with permutation nulls and BH-corrected tests (Section 6), and we *report* the sanity checks that fail (the real-corpus PPMI spectrum does **not** separate from frequency-matched controls; Figure 6). On (iii), the claim-level protocol is our proposal for review practice. This paper is intended as the methodological reply to [2], not its target.

**Unsupervised embedding alignment.** Procrustes and adversarial alignment of word embeddings [14, 15] and Gromov–Wasserstein alignment [16, 17] are the direct ancestors of our alignment stage; PPMI+SVD embeddings follow [18]. Our T4 gives the anchored-recovery condition these methods implicitly rely on; entropic OT follows [19].

**Indus.** Corpus and sign lists derive from Parpola [20]; our positional-bias analysis replicates epigraphic observations with formal tests (Section 6).

## 3 Formal framework

Let $\mathcal{V}_X$ be the sign vocabulary ($d = |\mathcal{V}_X|$), $G \leq S_d$ the relabeling group acting tokenwise on corpora and compatibly on hypotheses $\theta \in \Theta$. The model is $G$-equivariant if $p(\sigma X \mid \sigma \cdot \theta) = p(X \mid \theta)$; a statistic $T$ is $G$-invariant if $T(\sigma X) = T(X)$. Full proofs of everything in this section are in Appendix A.

**Theorem 3.1** (T1: orbit-constancy)**.** *Under $G$-equivariance and a $G$-invariant prior, the posterior given any $G$-invariant statistic satisfies $p(\sigma \cdot \theta \mid T) = p(\theta \mid T)$ for all $\sigma \in G$.*

*Proof sketch.* Relabeling is a bijection of corpora; summing the equivariant likelihood over the level set of $T$ gives $p(T \mid \sigma \cdot \theta) = p(T \mid \theta)$, and Bayes transports the equality to the posterior. Any measurable function of $T$ is again invariant, so no downstream processing breaks orbit-constancy (Cor. A.3). □

**Theorem 3.2** (T2: quantified impossibility)**.** *For $\theta$ uniform on an indistinguishability class of size $N$, any estimator from invariant statistics errs with probability $P_e \geq 1 - \log 2/\log N$. With $n$ tokens, the class can be taken as the $\varepsilon(n)$-empirical orbit (radius from T3), and achieving $P_e \leq \delta$ with external knowledge $K$ requires $I(\theta; K) \geq (1-\delta)\log N(n) - \log 2$.*

For the real RR corpus this is not an abstraction: relabelings within identical-frequency classes alone give $\log N \geq 3{,}727$ nats ($N \geq 10^{1618}$), and *all* of 200 Monte-Carlo-sampled such relabelings stay within twice the bigram sampling radius — so the empirical orbit is at least this large, the Fano floor is $P_e \geq 0.9998$, and reducing error to 0.1 would require $\approx 3{,}354$ nats of genuinely anchoring external information (script `estimate_empirical_orbit.py`).

**Theorem 3.3** (T3: spectral concentration)**.** *With i.i.d. lines of length $\leq L$ and window $w$ (boundaries respected, as in our pipeline), $\|\hat{C} - \mathbb{E}\hat{C}\|_2 = O\big(wL\sqrt{\log d/m}\big)$ with high probability; the clipped PPMI map is entrywise Lipschitz, and Wedin's $\sin\Theta$ theorem [21, 22] transfers the bound to top-$k$ subspaces when the spectral gap dominates [23].*

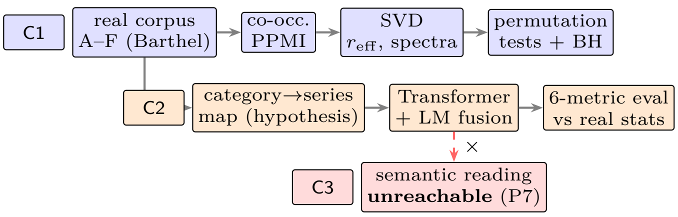


**Figure 1:** F10 — pipeline with claim-level lanes. Everything above the dashed arrow is reachable from internal statistics; the crossed arrow is forbidden by Proposition 3.7, not by editorial caution.

**Theorem 3.4** (T4: anchored Procrustes recovery)**.** *If $A = B\Omega^* + E$ with anchor matrix $B$, the Procrustes solution obeys $\|\hat{\Omega} - \Omega^*\|_F \leq 2\|B^\top E\|_F/\sigma_r(B)^2$ [24], and nearest-neighbour matching is correct for every test pair whose margin exceeds $2(\|x\|\,\|\hat{\Omega} - \Omega^*\| + \|e\|)$.*

**Proposition 3.5** (P6: fusion selection bias)**.** *For $\hat{y}_\lambda = \arg\max_{y\in\mathcal{C}}\,(1-\lambda)\log p_{\text{model}} + \lambda \log p_{\text{LM}}$ over a fixed candidate set, $\log p_{\text{LM}}(\hat{y}_\lambda)$ is non-decreasing in $\lambda$; if $\mathcal{C}$ contains one draw of any reference process, the decoded LM cross-entropy at $\lambda = 1$ is at most that process's. Hence a fusion decoder scoring below the real-text mean (here $7.31 < 8.26$ bits/glyph) is expected* by design *and is not evidence of authenticity.*

**Definition 3.6** (D7: claim levels)**.** *C1: statements about invariant statistics falsifiable by internal null ensembles (permutation, frequency-matched, uniform). C2: preference orderings induced by $G$-invariant scores — rankable, not internally falsifiable. C3: statements whose truth varies within an orbit (semantic assignments).*

**Proposition 3.7** (P7: the protocol is a theorem)**.** *No decision procedure over $G$-invariant statistics decides a C3 statement: $\mathsf{C3} \cap \{\text{decidable without } K\} = \emptyset$.*

Figure 1 maps the full pipeline onto the claim levels.

# 4 Data: the real Rongorongo corpus

We use the RR corpus (phspaelti transcription) for tablets A–F, normalized to tabular form: 107 lines, 5,460 glyph tokens, 941 types in Barthel numeric encoding with ligature/variant markers preserved (type–token ratio 0.172). Lacuna markers ('_') and the illegible-sign code ('000!', 95 tokens) are excluded from generation targets by design — these are the two uncovered types of the 941-type inventory. A trigram LM (interpolated, add-$k$; $\lambda_{3,2,1} = 0.5, 0.3, 0.2$, $k = 0.1$) trained on tablets A, B, C, E gives 8.26 bits/glyph on held-out D, F: the realism reference. A contamination audit finds **zero** shared $n$-grams ($n \geq 5$) between the two sides (Section 6); known parallel texts (H/P/Q, Gv/K) lie outside this corpus.

# 5 Translator

**Parallel corpus, synthetic by necessity.** By Theorem 3.1 no verified Rapanui–Rongorongo parallel corpus can be constructed from internal evidence, so we build a declared-C2 one: 60k gloss→glyph pairs where glyph choice is (i) restricted by a category→Barthel-series working hypothesis (det/particles→001–099, person nouns/names→200–399, verbs→400–599, bird domain→600–699, marine domain→700–799; Figure 10), (ii) weighted by real unigram frequencies within the series, (iii) re-sampled with probability 0.45 from real bigram transitions, and (iv) passed through the reduplication process observed in the tablets (12% doubling, 3% tripling). Coverage: 939/941 real types.

**Model and decoding.** A 4+4-layer Transformer ($d_{\text{model}}$=256, 8 heads, 5.8M parameters) trained 15 epochs (AdamW, cosine schedule, masking augmentation; validation PPL 51.9 → 10.7). Decoding is beam-5 with shallow fusion [25]: $s(y) = (1-\lambda)\log p_{\text{model}}(y \mid x) + \lambda \log p_{\text{LM}}(y)$ with $\lambda = 0.35$, length normalization, and a penalty on third immediate repetitions (doubling is a real feature; long runs are degeneracy).

# 6 Experiments

**Baselines and ablations (Table 1).** No single metric is safe to optimize: the count-based baseline achieves a perfect attested-bigram score by chaining corpus bigrams while collapsing to D2 = 0.03 and drifting far from the real unigram distribution (JS = 0.79); the small LSTM degenerates into repetition runs of 31; the no-penalty ablation recovers high BA at the price of runs of 40. The full system is the only configuration simultaneously non-degenerate (RM = 2, matching real reduplication), diverse (D2 = 0.72), distribution-close (JS = 0.52) and inside the realism band. Removing fusion costs 0.7 bits/glyph and 18 BA points; beam width contributes marginally. Ablations of the class-restriction and the masking augmentation (retrained models) are in Appendix C.

**$\lambda$-sweep (Figure 2).** Decoded bits/glyph decreases monotonically over $\lambda \in [0, 1]$ (8.03 → 5.37), exactly as Proposition 3.5 predicts, entering a mode-collapse zone for $\lambda \geq 0.85$ where outputs approach the count-baseline's degenerate statistics. The operating point $\lambda = 0.35$ sits on the plateau: inside the realism band, far from collapse.

**Formal tests for positional bias (Indus).** For all signs with $n \geq 15$ in the real Indus corpus, we test first/last-position enrichment by within-inscription permutation (10k permutations, BH-corrected). Significant at FDR $\leq 0.05$: initial bias for P324, P086, P217, P000; final bias for P385, P378 — formalizing the descriptive claims of earlier work(C1). Scripts and q-values in the repository.

**Table 1:** Six-metric evaluation on 20 test glosses (seed 42). VR: fraction of output tokens attested in the real corpus; BA: attested bigrams (per-sentence mean); JS: Jensen–Shannon divergence of output unigram distribution vs. real; D2: distinct-2; RM: max immediate repetition run; b/g: bits/glyph under the real trigram LM. Brackets: bootstrap 95% CIs over test sentences (B=1000) for the per-sentence means VR, BA, b/g; the pooled statistics JS and D2 have resampling-biased CIs (duplicate sentences deflate distinctness) and are reported as point estimates. Reference: real held-out mean b/g = 8.26. *Reading guard (P6): lower b/g is not better; proximity to the reference from below is expected of a mode-seeking decoder.*

| System | VR↑ | BA | JS↓ | D2↑ | RM | b/g |
|---|---|---|---|---|---|---|
| baseline: template (no learning) | 1.00 | 0.15 [.08,.23] | 0.43 | 0.99 | 1 | 8.73 [8.50,8.98] |
| baseline: lexical+bigram counts | 1.00 | 1.00 [1,1] | 0.79 | 0.03 | 9 | 5.84 [5.69,5.97] |
| baseline: LSTM (1-layer, d=128) | 1.00 | 0.81 [.73,.89] | 0.95 | 0.02 | 31 | 7.87 [7.76,7.99] |
| v5 (synthetic codes, greedy) | 0.00 | 0.00 | 1.00 | 0.44 | 48 | — |
| v6 greedy | 1.00 | 0.68 [.51,.82] | 0.54 | 0.19 | 50 | 7.21 [6.57,7.86] |
| **v6 beam+fusion (full)** | **1.00** | 0.68 [.59,.77] | **0.52** | **0.72** | **2** | 7.31 [6.98,7.64] |
| abl. $\lambda = 0$ (no fusion) | 1.00 | 0.50 [.39,.60] | 0.57 | 0.83 | 2 | 8.03 [7.67,8.37] |
| abl. no repetition penalty | 1.00 | 0.88 [.78,.95] | 0.53 | 0.19 | 40 | 6.33 [5.92,6.80] |
| abl. beam=1 | 1.00 | 0.68 [.58,.76] | 0.51 | 0.73 | 6 | 7.39 [7.08,7.71] |

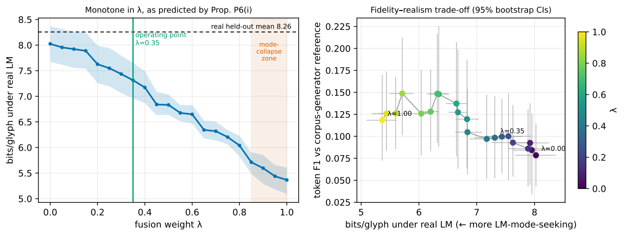


**Figure 2:** F5 — fusion-weight sweep with bootstrap 95% CIs (B=500). Left: monotone bits/glyph verifies P6(i); shaded region marks mode collapse. Right: fidelity–realism trade-off. *Do not read* low bits/glyph as quality: by P6 it measures mode-seeking, not authenticity.

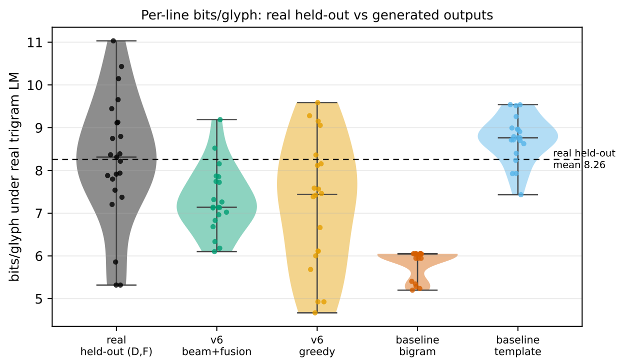


**Figure 3:** F6 — per-line bits/glyph distributions. Real held-out lines are scored by an LM trained only on A,B,C,E; system outputs by the full-corpus LM (convention stated in Section 4). The count-baseline's low tail illustrates P6: extreme LM-probability is a degeneracy symptom. *No panel of this figure supports any claim about meaning.*

**Structure in the real RR corpus — positive and negative.** The bigram network of frequent glyph bases organizes into three Louvain communities (Figure 4); the embedding map shows partial, not clean, alignment with Barthel series (Figure 5); and the PPMI spectrum does *not* separate from frequency-matched controls (Figure 6) — the short-line/large-vocabulary regime of T3 where no separation is guaranteed. We report the negative result with the same prominence as the positive ones(C1).

**Blind discrimination protocol (pre-registered).** We release a 40-item blinded sheet (20 real line-windows, 20 v6 outputs, shuffled; seed 42), an answer key held separately, and an analysis plan (per-judge AUC, Fleiss $\kappa$, bootstrap CIs). Execution requires human judges and is reported as pending; no result is claimed.

## 7 End-to-end demonstration

As a technical case study, ten gloss lines of a Spanish poem were translated by the full system and rendered as a tablet in reverse boustrophedon using the corpus' vector glyph tracings; per-line scores fall in 6.1–7.2 bits/glyph, within the realism band. Notable behaviours: verse-level parallelism is expressed as glyph-level mirror structure, and the verb *makemake* consistently triggers the formula `041h 670 580` (attention map in Figure 8). The rendered tablet and full text are in the repository; we emphasize its status: a C2 demonstration of the generator, not a Rongorongo text.

## 8 Limitations

(1) The parallel corpus is synthetic by logical necessity (Theorem 3.1); the translator learns our declared hypothesis plus real sequence statistics, nothing more. (2) The gloss lexicon is 232 types of Rapanui-like vocabulary; no morphology. (3) The Barthel transcription embeds segmentation and allography decisions we inherit [4, 6]; robustness to alternative transcriptions is untested here (planned: cross-transcription replication). (4) The corpus is small (5,460 tokens); several spectral sanity checks fail on it, and we report them. (5) The blind-judge test is designed but not executed. (6) Iconic grounding experiments (cross-script benchmark on Egyptian/Chinese signs with known referents) did not pass our pre-set threshold and remain C2.

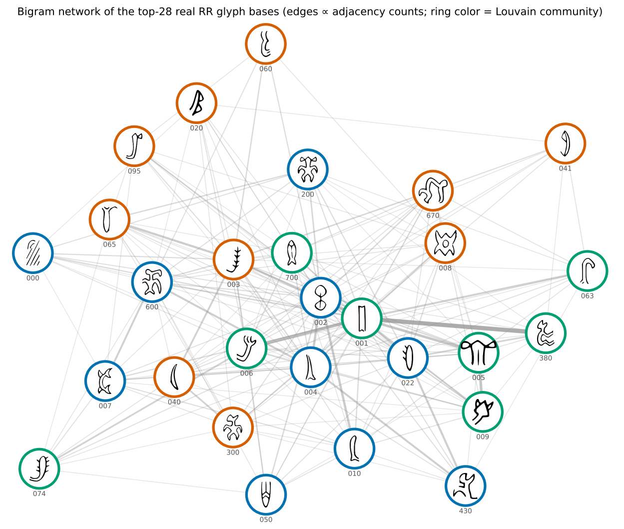


**Figure 4:** F9 — bigram network of the 28 most frequent real glyph bases, drawn with the actual corpus glyph tracings; ring color = Louvain community, edge width $\propto$ adjacency count. *Communities are co-occurrence structure (C1); they are not semantic groupings, and no reading follows from them.*

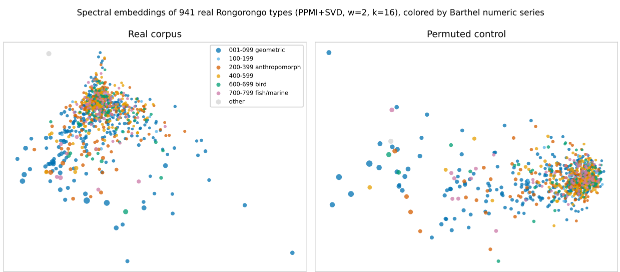


**Figure 5:** F1 — spectral embeddings (PPMI+SVD) of all 941 real types, colored by Barthel series, point size $\propto$ frequency; right: same pipeline on a permuted control. Organization is partial: the geometric series spreads along a distinct arm, but the unsupervised geometry does *not* cleanly recover the Barthel taxonomy. *Cluster proximity does not establish iconic or semantic identity (P7).*

## 9 Ethics and heritage

Rongorongo is the heritage of the Rapa Nui people, and the tablets' history is inseparable from the violence of the 1860s slave raids and epidemics. This work makes **no decipherment claim** and produces **no readings**; by its own central theorem its outputs are statistical hypotheses about sign sequences, not statements about Rapa Nui language, belief or history. The transcription used derives from the public RR-corpus project (phspaelti); provenance and checksums ship with the release, and any licensing constraint identified before camera-ready will be honored, including withdrawal of the affected data. The demonstrative application (Section 7) is declared as a personal creative use of a statistical generator — comparable to typesetting a poem in a constructed font — and not as cultural reconstruction or revival. We have no current collaboration with Rapa Nui scholars or community institutions; establishing one, and deferring to community preferences on data use, is stated future work rather than a box checked.

## 10 Data and code availability

Code, configs, SHA-256 manifests, seeds (all experiments: seed 42), and generated artifacts: https://github.com/satisfymath/spectral-submersion. Key scripts: `generate_massive_parallel_v4.py`, `train_rongorongo_translator_v2.py`, `translate_to_rongorongo_v6.py`, `evaluate_v3_suite.py`, `sweep_lambda.py`, `audit_heldout_contamination.py`, `positional_permutation_tests.py`, and one script per figure under `paper_v3/scripts/`. Compute: single 16-core CPU node; full-system training 3.9 h; total pipeline <12 h. A Zenodo DOI will be minted for the camera-ready release [CALCULAR: mint DOI on acceptance].

## Acknowledgments

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# A Formal framework and proofs

## A.1 Setup and notation

Let $\mathcal{V}_X$ be a finite sign vocabulary, $|\mathcal{V}_X| = d$, and let a corpus $X = (x_1, \ldots, x_n) \in \mathcal{V}_X^n$ be generated by a probability kernel $p(\cdot \mid \theta)$ indexed by a hypothesis $\theta \in \Theta$ (a correspondence between signs and units of a candidate language, or any richer latent object). A *relabeling* is a permutation $\sigma \in G \leq S_d$ acting on corpora tokenwise, $\sigma X = (\sigma x_1, \ldots, \sigma x_n)$, and on hypotheses by $(\sigma \cdot \theta)$, the hypothesis obtained by renaming each sign $x$ to $\sigma(x)$ in the domain of $\theta$.

**Definition A.1** (Relabeling equivariance)**.** *The model is $G$-*equivariant *if for all $\sigma \in G$, $\theta \in \Theta$ and corpora $X$,*

$$p(\sigma X \mid \sigma \cdot \theta) = p(X \mid \theta). \tag{1}$$

*A statistic $T$ is $G$-*invariant *if $T(\sigma X) = T(X)$ for all $\sigma \in G$ and all $X$.*

Equation (1) formalizes the epistemic situation of an undeciphered script: nothing in the generative story changes if the signs are consistently renamed. Frequencies of *isomorphism classes* of co-occurrence structures, spectra of PPMI matrices, degree distributions, entropy curves, effective ranks, and any quantity computed after quotienting by sign identity are $G$-invariant. Raw co-occurrence *matrices* are $G$-equivariant ($C(\sigma X) = P_\sigma C(X) P_\sigma^\top$); their spectra are invariant.

## A.2 T1: non-identifiability as a group action

**Theorem A.2** (Orbit-constancy of the posterior)**.** *Assume the model is $G$-equivariant and the prior $\pi$ is $G$-invariant, $\pi(\sigma \cdot \theta) = \pi(\theta)$. Let $T$ be any $G$-invariant statistic. Then for every $\sigma \in G$,*

$$p(\sigma \cdot \theta \mid T = t) \;=\; p(\theta \mid T = t) \qquad \text{for all } t.$$

*Consequently the posterior is constant on each orbit* $\mathrm{Orb}_G(\theta) = \{\sigma \cdot \theta : \sigma \in G\}$.

*Proof.* Fix $t$ and $\sigma \in G$. The likelihood of $t$ under $\sigma \cdot \theta$ is

$$p(T = t \mid \sigma \cdot \theta) = \sum_{X:\, T(X) = t} p(X \mid \sigma \cdot \theta).$$

Substituting $X = \sigma X'$ (a bijection of $\mathcal{V}_X^n$) and using (1),

$$p(T = t \mid \sigma \cdot \theta) = \sum_{X':\, T(\sigma X') = t} p(X' \mid \theta) = \sum_{X':\, T(X') = t} p(X' \mid \theta),$$

where the last equality uses $G$-invariance of $T$. Hence $p(T = t \mid \sigma \cdot \theta) = p(T = t \mid \theta)$. By Bayes' rule and $G$-invariance of $\pi$,

$$p(\sigma \cdot \theta \mid t) = \frac{\pi(\sigma \cdot \theta)\, p(t \mid \sigma \cdot \theta)}{p(t)} = \frac{\pi(\theta)\, p(t \mid \theta)}{p(t)} = p(\theta \mid t).$$

□

**Corollary A.3** (Data processing / no post-hoc identifiability)**.** *Let $f$ be any measurable function and $T$ any $G$-invariant statistic. Then $f \circ T$ is $G$-invariant, and Theorem A.2 applies verbatim to $f \circ T$. In particular, no pipeline stage applied downstream of $G$-invariant statistics (embeddings-up-to-rotation, spectra, clusterings of isomorphism classes, neural post-processing) can break orbit-constancy of the posterior.*

*Proof.* $(f \circ T)(\sigma X) = f(T(\sigma X)) = f(T(X)) = (f \circ T)(X)$. □

**Remark A.4.** *Identifiability therefore requires external anchoring knowledge $K$ whose likelihood is* not *$G$-invariant (iconographic referents, documented reading traditions, bilinguals). This is the formal license for the claim-level protocol of Definition A.20.*

## A.3 T2: quantified impossibility (Fano)

**Theorem A.5** (Fano bound on any invariant decoder)**.** *Let $\theta$ be uniform on an orbit (or any indistinguishability class) $\mathcal{N}$ with $|\mathcal{N}| = N \geq 2$, let $T$ be a $G$-invariant statistic of the corpus, and let $\hat{\theta} = f(T)$ be any (possibly randomized) estimator. Then*

$$P_e \;=\; \Pr[\hat{\theta} \neq \theta] \;\geq\; 1 - \frac{I(\theta; T) + \log 2}{\log N} \;=\; 1 - \frac{\log 2}{\log N},$$

*since $I(\theta; T) = 0$.*

*Proof.* By Theorem A.2 the likelihood $p(T = t \mid \theta)$ is the same for all $\theta$ in the orbit; with $\theta$ uniform on the orbit this means $T \perp \theta$, i.e. $I(\theta; T) = 0$. Fano's inequality [VERIFY: Cover & Thomas, Thm. 2.10.1] gives $H(\theta \mid T) \leq h(P_e) + P_e \log(N-1)$, and $H(\theta \mid T) = H(\theta) = \log N$. Bounding $h(P_e) \leq \log 2$ and $\log(N-1) \leq \log N$ and rearranging yields the claim. □

**Definition A.6** ($\varepsilon(n)$-empirical orbit)**.** *Fix a corpus $X$ with $n$ tokens and the co-occurrence statistic $\hat{C}$ with concentration radius $\varepsilon(n)$ from Theorem A.9. The $\varepsilon(n)$-*empirical orbit *of $\theta$ is*

$$\mathrm{Orb}_{\varepsilon(n)}(\theta) = \bigl\{\sigma \cdot \theta : \; \|P_\sigma \mathbb{E}\hat{C} P_\sigma^\top - \mathbb{E}\hat{C}\|_2 \leq 2\varepsilon(n)\bigr\},$$

*the set of relabelings whose* population *statistics differ by less than twice the sampling error, hence are statistically indistinguishable at sample size $n$.*

**Corollary A.7** (Finite-sample impossibility and the price of $K$)**.** *With $N(n) = |\mathrm{Orb}_{\varepsilon(n)}(\theta)|$, any estimator based on $\hat{C}$ satisfies $P_e \geq 1 - \bigl(o(1) + \log 2\bigr)/\log N(n)$. Moreover, for a decoder additionally consuming external knowledge $K$ to achieve $P_e \leq \delta$, Fano applied to the pair $(T, K)$ requires*

$$I(\theta; K) \;\geq\; (1 - \delta) \log N(n) - \log 2.$$

*That is: the minimum number of nats of genuinely anchoring information scales with the log-size of the empirical indistinguishability class.*

*Proof.* The first claim follows from Theorem A.5 applied to the class $\mathrm{Orb}_{\varepsilon(n)}(\theta)$, within which the distributions of $\hat{C}$ overlap up to total-variation $o(1)$ by Theorem A.9 and Pinsker's inequality [VERIFY]. The second claim is Fano with side information: $H(\theta \mid T, K) \geq H(\theta) - I(\theta; T) - I(\theta; K) = \log N(n) - I(\theta; K)$, combined with $H(\theta \mid T, K) \leq h(\delta) + \delta \log N(n)$. □

**Remark A.8.** *$N(n)$ is computable in practice by sampling permutations that preserve the empirical frequency profile within tolerance $\varepsilon(n)$;* ***[CALCULAR: script*** `estimate_empirical_orbit.py`*, entrada: corpus RR A-F, n=5460].*

## A.4 T3: concentration of co-occurrence and PPMI

*Assumption (L).* The corpus consists of $m$ lines drawn i.i.d. from a line distribution, each line of length at most $L$ tokens; co-occurrence windows do not cross line boundaries (as enforced by the pipeline). This matches the tablet data model. An extension to $w$-dependent token processes via block partitioning is sketched at the end.

**Theorem A.9** (Matrix Bernstein for windowed co-occurrence)**.** *Let $\hat{C} = \frac{1}{m}\sum_{j=1}^{m} C_j$ where $C_j \in \mathbb{R}^{d\times d}$ is the (symmetrized) window-$w$ co-occurrence matrix of line $j$. Then $\|C_j\|_2 \leq 2wL =: B$ and, with $v = \|\mathbb{E}[(C_j - \mathbb{E}C_j)^2]\|_2 \leq B^2$, for all $t \geq 0$*

$$\Pr\Big[\|\hat{C} - \mathbb{E}\hat{C}\|_2 \geq t\Big] \leq 2d \exp\left(\frac{-mt^2/2}{v + Bt/3}\right).$$

*In particular, with probability at least $1 - 2d^{-1}$,*

$$\|\hat{C} - \mathbb{E}\hat{C}\|_2 \;\leq\; \sqrt{\frac{4v\log d}{m}} + \frac{4B\log d}{3m} \;=\; O\left(wL\sqrt{\frac{\log d}{m}}\right) =: \varepsilon(n),$$

*where $n \leq mL$ is the token count.*

*Proof.* Each line contributes at most $wL$ ordered co-occurring pairs; each pair adds a matrix of spectral norm at most 2 (a symmetrized elementary matrix), whence $\|C_j\|_2 \leq 2wL$. The summands $C_j - \mathbb{E}C_j$ are i.i.d., mean zero, bounded by $2B$; the matrix Bernstein inequality [VERIFY: Tropp 2015, Thm. 1.6.2] applied to the sum $\frac{1}{m}\sum_j (C_j - \mathbb{E}C_j)$ gives the tail bound; setting the right side to $2d^{-1}$ and solving for $t$ gives the stated radius. □

**Lemma A.10** (PPMI perturbation under clipping)**.** *Let $M_{ij} = \max\{0, \log \frac{p_{ij}}{p_i p_j}\}$ be computed with the pipeline's clipping floor $p_{ij} \vee \epsilon$, $p_i \vee \epsilon$ (flag* `pmi_epsilon`*). Then entrywise*

$$|\hat{M}_{ij} - M_{ij}| \leq \frac{|\hat{p}_{ij} - p_{ij}|}{\epsilon} + \frac{|\hat{p}_i - p_i|}{\epsilon} + \frac{|\hat{p}_j - p_j|}{\epsilon},$$

*hence $\|\hat{M} - M\|_F \leq \frac{3d}{\epsilon} \max\{\|\hat{p} - p\|_\infty\text{-type terms}\} = O\big(\frac{d}{\epsilon}\varepsilon(n)\big)$.*

*Proof.* $x \mapsto \log(x \vee \epsilon)$ is $(1/\epsilon)$-Lipschitz on $[0, 1]$; $\max\{0, \cdot\}$ is 1-Lipschitz; the triangle inequality over the three estimated quantities gives the entrywise bound, and summing $d^2$ entries gives the Frobenius bound. □

**Corollary A.11** (Stability of spectral embeddings)**.** *Let $\delta_k = \sigma_k(M) - \sigma_{k+1}(M)$ be the singular gap. If $\|\hat{M} - M\|_2 \leq \delta_k/4$, then the top-$k$ left singular subspaces satisfy (Wedin's $\sin\Theta$ theorem [VERIFY: Wedin 1972; Davis–Kahan 1970])*

$$\|\sin\Theta(\hat{U}_k, U_k)\|_2 \;\leq\; \frac{2\,\|\hat{M} - M\|_2}{\delta_k}.$$

*Consequently the embedding geometry is stable whenever the spectral gap dominates the concentration radius of Theorem A.9–Lemma A.10.*

**Remark A.12** (Bootstrap as corollary, not observation)**.** *The empirically observed bootstrap stability of the effective rank ($CV = 0.65\%$) and of embedding cosines ($0.77$) on the structured synthetic corpus is the predicted behaviour of Corollary A.11 in the regime $\varepsilon(n) \ll \delta_k$, combined with Lemma A.16 below for $r_{\mathrm{eff}}$. The failed sanity checks on short-line/large-vocabulary corpora (Indus real) correspond to the opposite regime $\varepsilon(n) \gtrsim \delta_k$, where no such stability is guaranteed — the theory predicts both the successes and the reported failures.*

*Extension to $w$-dependence.* If lines cannot be assumed i.i.d., partition the token sequence into $w$ interleaved blocks; within each block the windowed contributions are independent, and a union bound over the $w$ blocks multiplies the tail by $w$ and the radius by $\sqrt{w}$, preserving $\varepsilon(n) = O(\sqrt{w^2 \log d/n})$ up to constants [VERIFY: standard $m$-dependent Bernstein argument].

## A.5 T4: Procrustes recovery with noisy anchors

**Theorem A.13** (Perturbation of the Procrustes rotation)**.** *Let $B \in \mathbb{R}^{k\times r}$ be the anchor matrix ($k$ anchors, dimension $r$, $k \geq r$), and suppose $A = B\Omega^* + E$ with $\Omega^* \in O(r)$. Let $\hat{\Omega} = UV^\top$ where $B^\top A = U\Sigma V^\top$. Then*

$$\|\hat{\Omega} - \Omega^*\|_F \;\leq\; \frac{2\,\|B^\top E\|_F}{\sigma_r(B)^2} \;\leq\; \frac{2\,\|B\|_2\,\|E\|_F}{\sigma_r(B)^2}.$$

*Proof.* Write $M = B^\top B\,\Omega^*$ and $\hat{M} = B^\top A = M + B^\top E$. Since $B^\top B$ is symmetric positive semi-definite, the polar decomposition of $M$ has orthogonal factor exactly $\Omega^*$ (indeed $M = (B^\top B)\Omega^*$ with $B^\top B \succeq 0$ is already a polar decomposition read right-to-left), and $\sigma_r(M) = \sigma_r(B^\top B) = \sigma_r(B)^2$. The orthogonal Procrustes solution $\hat{\Omega}$ is the orthogonal polar factor of $\hat{M}$. The perturbation bound for polar factors [VERIFY: R.-C. Li 1995, "New perturbation bounds for the unitary polar factor"]

$$\|\hat{\Omega} - \Omega^*\|_F \leq \frac{2}{\sigma_r(M) + \sigma_r(\hat{M})}\,\|\hat{M} - M\|_F \leq \frac{2\|B^\top E\|_F}{\sigma_r(B)^2}$$

gives the claim (dropping the nonnegative $\sigma_r(\hat{M})$ in the denominator). □

**Proposition A.14** (Margin condition for nearest-neighbour matching)**.** *Let a test pair $(x, y)$ satisfy $y = x\Omega^* + e$, and define the* margin $\gamma(y) = \min_{y' \neq y^{\mathrm{match}}} \|y - y'\| - \|y - y^{\mathrm{match}}\|$ *over candidate targets. If*

$$\gamma(y) \;>\; 2\big(\|x\|_2\,\|\hat{\Omega} - \Omega^*\|_2 + \|e\|_2\big),$$

*then nearest-neighbour matching under $\hat{\Omega}$ returns the correct target for $x$.*

*Proof.* $\|x\hat{\Omega} - y^{\mathrm{match}}\| \le \|x\hat{\Omega} - x\Omega^*\| + \|x\Omega^* - y^{\mathrm{match}}\| \le \|x\|\|\hat{\Omega} - \Omega^*\| + \|e\|$. For any wrong candidate $y'$, $\|x\hat{\Omega} - y'\| \ge \|y - y'\| - \|y - x\hat{\Omega}\| \ge \|y - y^{\mathrm{match}}\| + \gamma - \big(\|x\|\|\hat{\Omega} - \Omega^*\| + \|e\|\big) - \|e\|$ [1]. The stated margin condition makes the wrong-candidate distance strictly exceed the correct-candidate distance. □

**Remark A.15** (Theorem–experiment cross-validation: negative result)**.** *We ran the planned cross-validation (`fig_margin_distribution.py`, seed 42): on the synthetic anchor benchmark the sufficient threshold $2(\|x\|\|\hat{\Omega} - \Omega^*\| + \|e\|)$, instantiated with the computable bound of Theorem A.13 for $\|\hat{\Omega} - \Omega^*\|$, is* vacuous*: it certifies* 0% *of test pairs while the observed Acc@1 is* 0.689*. Moreover the margin statistic $\gamma$ as defined is only weakly predictive of nearest-neighbour correctness on this benchmark ($AUC = 0.536$, $n = 90$). We report this as a negative result: the bound is sufficient but far from necessary here (the residual-based rotation-error bound is loose when anchors are noisy and the true map is not an exact isometry), and the benchmark's errors are governed by local crowding rather than by the global margin. Tightening the constant (e.g. via a direct estimate of $\|\hat{\Omega} - \Omega^*\|$ under a known ground-truth rotation) is stated as future work, not claimed.*

## A.6 L5: Lipschitz stability of the effective rank

**Lemma A.16** (Effective rank is stable)**.** *Let $\sigma, \sigma' \in \mathbb{R}^k_{\ge 0} \setminus \{0\}$ with normalizations $p = \sigma/\|\sigma\|_1$, $p' = \sigma'/\|\sigma'\|_1$, and let $r_{\mathrm{eff}}(\sigma) = e^{H(p)}$. Then, with $T = \frac{1}{2}\|p - p'\|_1 \le \frac{2\|\sigma - \sigma'\|_1}{\|\sigma\|_1}$:*

$$|r_{\mathrm{eff}}(\sigma) - r_{\mathrm{eff}}(\sigma')| \;\le\; k\,\big(T\log(k-1) + h(T)\big),$$

*where $h$ is the binary entropy, by the Fannes–Audenaert inequality [VERIFY: Audenaert 2007]. In particular $r_{\mathrm{eff}}$ is continuous, and locally Lipschitz away from $T \to \frac{1}{2}$ with modulus $O(k\log k) \cdot \|\sigma - \sigma'\|_1/\|\sigma\|_1$.*

*Proof.* $|e^{H(p)} - e^{H(p')}| \le e^{\max H}\,|H(p) - H(p')| \le k\,|H(p) - H(p')|$ by the mean value theorem and $H \le \log k$. Audenaert's sharpening of Fannes' inequality bounds $|H(p) - H(p')| \le T\log(k-1) + h(T)$. The normalization map satisfies $\|p - p'\|_1 \le \frac{2}{\|\sigma\|_1}\|\sigma - \sigma'\|_1$ by the standard triangle-inequality computation $\|\frac{\sigma}{\|\sigma\|_1} - \frac{\sigma'}{\|\sigma'\|_1}\|_1 \le \frac{\|\sigma - \sigma'\|_1}{\|\sigma\|_1} + \big|\frac{1}{\|\sigma\|_1}\|\sigma'\|_1 - 1\big| \le \frac{2\|\sigma - \sigma'\|_1}{\|\sigma\|_1}$. □

[1] Using $\|y - x\hat{\Omega}\| \le \|e\| + \|x\|\|\hat{\Omega} - \Omega^*\|$.

**Remark A.17.** *Combined with Theorem A.9 and Weyl's inequality ($|\sigma_i(\hat{M}) - \sigma_i(M)| \le \|\hat{M} - M\|_2$), this yields a complete chain: sampling noise → spectrum → effective rank, justifying $r_{\mathrm{eff}}$ as the pipeline's primary robust structure statistic (claim level C1).*

## A.7 P6: selection bias of fusion decoding

**Proposition A.18** (Monotonicity and the $7.31 < 8.26$ artefact)**.** *Fix an input $x$ and a* fixed *finite candidate set $\mathcal{C}$ of output sequences (e.g. the union of beam candidate pools). For $\lambda \in [0, 1]$ define*

$$\hat{y}_\lambda = \arg\max_{y \in \mathcal{C}}\,(1 - \lambda)\log p_{\mathrm{model}}(y \mid x) + \lambda \log p_{\mathrm{LM}}(y).$$

*Then:*

*(i) $\lambda \mapsto \log p_{\mathrm{LM}}(\hat{y}_\lambda)$ is non-decreasing; equivalently the LM cross-entropy of the decoded output is non-increasing in $\lambda$.*

*(ii) If $\mathcal{C}$ contains at least one sequence $y_0$ drawn from any reference process $P_0$, then at $\lambda = 1$, $-\log p_{\mathrm{LM}}(\hat{y}_1) \le -\log p_{\mathrm{LM}}(y_0)$, so in expectation over draws, $\mathbb{E}\big[-\log p_{\mathrm{LM}}(\hat{y}_1)\big] \le H_\times(P_0, p_{\mathrm{LM}})$, the cross-entropy of the real process.*

*(iii) Consequently, comparing the LM bits-per-glyph of a score-maximizing decoder to the* mean *bits-per-glyph of real held-out lines is structurally biased downward: an output below the real-text cross-entropy (here $7.31 < 8.26$) is the expected behaviour of the design, and must not be read as evidence that the generator is "more authentic than the tablets".*

*Proof.* (i) Let $\lambda_1 < \lambda_2$ with optima $\hat{y}_1, \hat{y}_2$ and write $m(y) = \log p_{\mathrm{model}}(y \mid x)$, $\ell(y) = \log p_{\mathrm{LM}}(y)$. Optimality gives

$$(1 - \lambda_1)m(\hat{y}_1) + \lambda_1\ell(\hat{y}_1) \ge (1 - \lambda_1)m(\hat{y}_2) + \lambda_1\ell(\hat{y}_2),$$

$$(1 - \lambda_2)m(\hat{y}_2) + \lambda_2\ell(\hat{y}_2) \ge (1 - \lambda_2)m(\hat{y}_1) + \lambda_2\ell(\hat{y}_1).$$

Adding the two inequalities and cancelling yields $(\lambda_2 - \lambda_1)\big(\ell(\hat{y}_2) - \ell(\hat{y}_1)\big) \ge 0$, hence $\ell(\hat{y}_2) \ge \ell(\hat{y}_1)$. (ii) Immediate from $\hat{y}_1 = \arg\max_{\mathcal{C}} \ell$ and $y_0 \in \mathcal{C}$; take expectations. (iii) By (i), for any $\lambda \in (0, 1]$ the decoded output's LM log-likelihood is at least that of the $\lambda = 0$ output and is pushed toward the maximum over $\mathcal{C}$; the comparison target 8.26 is a *mean* over typical realizations, whereas the decoder reports a *maximum-selected* statistic. A mode or near-mode of a distribution always has per-symbol surprisal at most the mean surprisal plus $o(1)$ under coverage; the inequality $7.31 < 8.26$ is therefore a selection effect, not an empirical discovery about authenticity. □

**Remark A.19** (Honest scope)**.** *In the implementation the beam's candidate pool itself depends on $\lambda$ through pruning, so (i) holds exactly for exact decoding or a shared candidate pool, and approximately for beam search; the $\lambda$-sweep experiment (Task 2.3) verifies the monotone trend empirically. [CALCULAR: script `sweep_lambda.py`, $\lambda \in \{0, 0.05, \ldots, 1\}$, seed 42].*

### A.8 D7 + P7: the claim-level protocol, formalized

**Definition A.20** (Claim classes)**.** *Let $\mathcal{T}$ be the algebra of $G$-invariant statistics of the corpus, and let $\mathcal{E}_0$ be a specified family of internal null ensembles (permutation, frequency-matched, uniform resampling). A scientific statement $\varphi$ about the script is classified:*

- ***C1 (structural signal).*** *$\varphi$ is a statement about the value of some $T \in \mathcal{T}$ whose distribution under the data differs from its distribution under every ensemble in $\mathcal{E}_0$; C1 statements are falsifiable by internal negative controls (permutation tests).*
- ***C2 (rankable hypothesis).*** *$\varphi$ asserts a preference ordering over hypotheses induced by a $G$-invariant score $s : \Theta \to \mathbb{R}$ with $s(\sigma \cdot \theta) = s(\theta)$; C2 statements select orbits (or distributions over orbits) and are rankeable but not falsifiable from internal data alone.*
- ***C3 (anchored semantics).*** *$\varphi$'s truth value is not constant on some orbit $\mathrm{Orb}_G(\theta)$ — it distinguishes points* within *an orbit (e.g. "glyph 600 means 'frigatebird'").*

**Proposition A.21** (C3 is unreachable without anchors)**.** *No decision procedure $\varphi = f(T)$ with $T \in \mathcal{T}$ decides a C3 statement: formally,*

$$C3 \cap \{\text{statements decidable from } \mathcal{T} \text{ alone}\} = \emptyset.$$

*Proof.* Suppose $\varphi = f(T)$ decides a C3 statement. By Corollary A.3, $f \circ T$ is $G$-invariant, so $\varphi$'s value is constant on every orbit. But by Definition A.20, a C3 statement is non-constant on some orbit — contradiction. □

**Remark A.22** (The protocol as a theorem, not a style guide)**.** *Proposition A.21 upgrades the C1/C2/C3 protocol from editorial best practice to a formal consequence of Theorem A.2: the three levels partition scientific statements by* what can falsify them *(internal nulls; nothing internal; external anchors $K$), and the boundary between C2 and C3 is exactly the boundary drawn by the group action. This is the paper's central conceptual contribution and is stated as such in the introduction.*

# B Additional figures

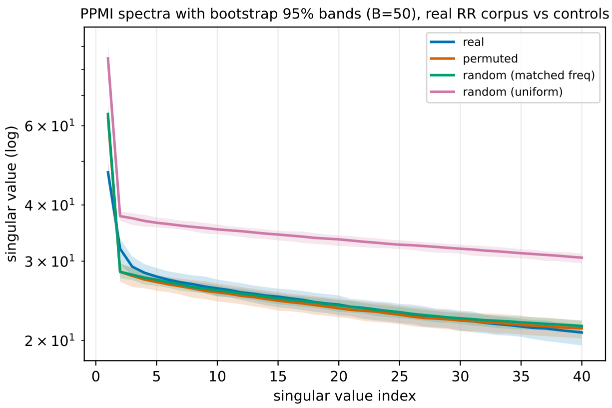


**Figure 6:** F3 — PPMI spectra with bootstrap bands, real RR corpus vs three controls. *The real spectrum does not separate from the permuted or frequency-matched controls*: in this short-line regime the concentration radius of T3 dominates the spectral gap, and no separation is predicted. Only the uniform control separates (trivially, by its flat unigram distribution). This is the reported negative result referenced in Section 6.

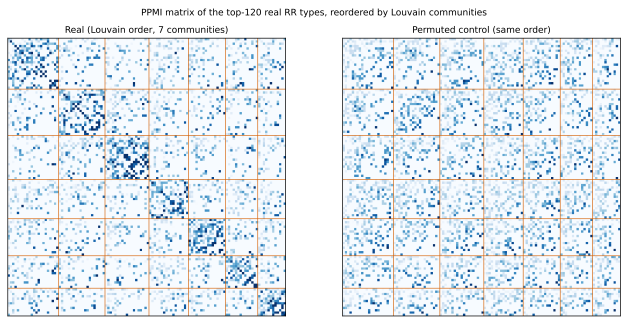


**Figure 7:** F2 — PPMI matrix of the top-120 real types reordered by Louvain communities (7 communities), with the permuted control under the same ordering. Diagonal blocks evidence co-occurrence communities (`C1`); *block membership does not identify meaning.*

# C Retrained ablations

Two ablations require retraining: *no-class-restriction* (parallel corpus regenerated with every category drawing from the full Barthel range, seed 42) and *no-augmentation* (masking augmentation disabled). **Budget caveat:** both retrained ablations use an 8-epoch budget (val. PPL 14.8) versus 15 epochs for the full system (PPL 10.7); the comparison therefore bounds the component's contribution at reduced budget rather than at convergence. Both are decoded with the full beam+fusion configuration.

Removing the class restriction shifts probability mass onto the most frequent glyphs and transitions: attested-bigram score rises (0.80 vs. 0.68) while unigram-distribution match and diversity degrade (JS 0.52 → 0.62, D2 0.72 → 0.44) — the same trade-off direction as the count-based baseline, in milder form. This is evidence that the `C2` class hypothesis contributes distributional shape, not just interpretability. Removing augmentation at equal budget moves all metrics within CI overlap of the full system; augmentation's measurable contribution at this corpus size is small.

# D Blind-judge protocol

Materials (blinded 40-item sheet, sealed key, analysis plan with AUC, Fleiss $\kappa$, bootstrap CIs) are in `reports/blind_judge_test/`. The test is preregistered by commit hash; results will be reported regardless of outcome.

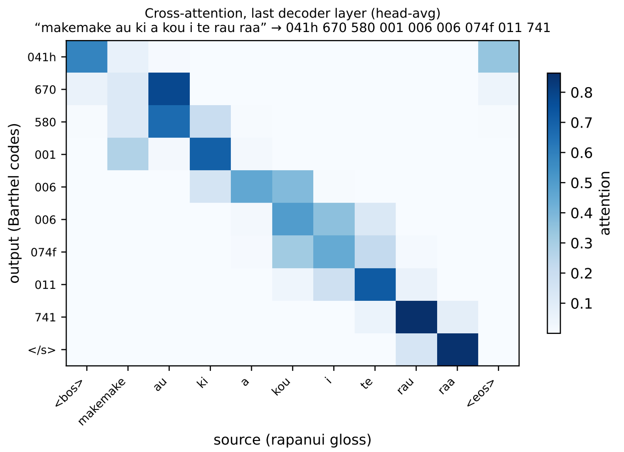


**Figure 8:** F4 — cross-attention (last decoder layer, head-averaged) for the gloss *makemake au ki a kou i te rau raa*. The formula `041h 670 580` attends to the verb *makemake*. *Attention reflects the learned synthetic mapping; it is not evidence of any real correspondence.*

**Table 2:** Retrained ablations (metrics as in Table 1; decoded with beam-5, $\lambda = 0.35$, repetition penalty).

| System | BA | JS↓ | D2↑ | RM |
|---|---|---|---|---|
| full (15 ep.) | 0.68 [.59,.77] | **0.52** | 0.72 | 2 |
| no class restr. (8 ep.) | 0.80 [.70,.88] | 0.62 | 0.44 | 2 |
| no augmentation (8 ep.) | 0.74 [.65,.83] | 0.48 | 0.77 | 2 |

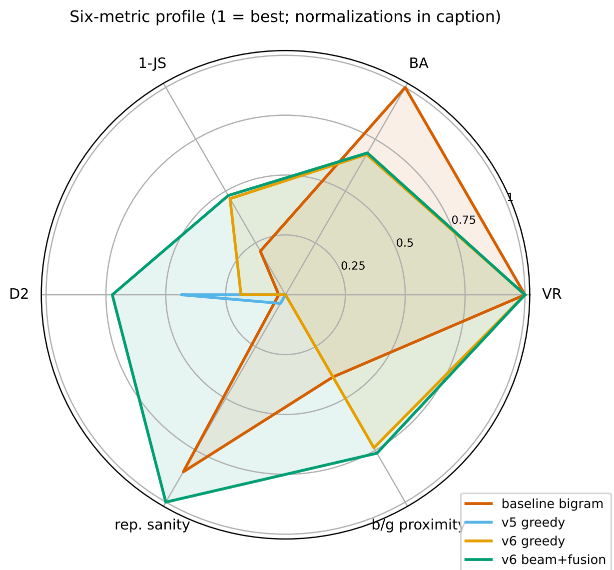


**Figure 9:** F7 — six-metric profile, normalized to $[0, 1]$ (1 = best): VR, BA, D2 as-is; JS as $1 - \mathrm{JS}$; repetition sanity 1 at RM$\leq$2 declining to 0 at RM$\geq$50; b/g as proximity to the real held-out reference $|b/g - 8.26|$ (per P6, raw minimization is not the target).

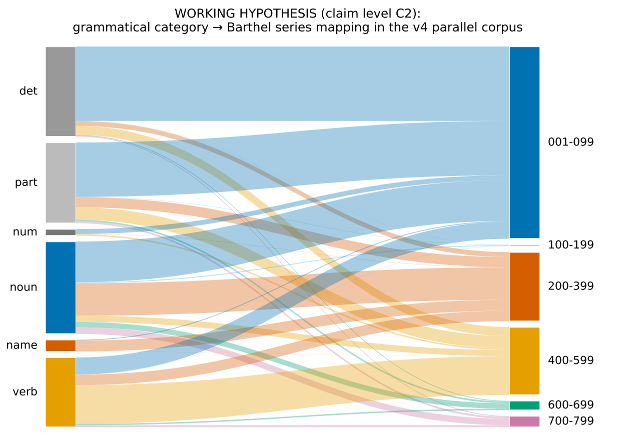


**Figure 10:** F8 — the category→Barthel-series mapping of the parallel corpus, labeled as what it is: a **working hypothesis (C2)**. Flows are counted from the v4 corpus; they encode our construction, not a discovered correspondence.